

Transfer Learning and Fine-Tuning

Frozen Extraction · End-to-End · Gradual Unfreezing · Discriminative LRs · Domain Shift



The Problem

You need to classify chest X-rays for pneumonia.
You have 2,000 labelled images.

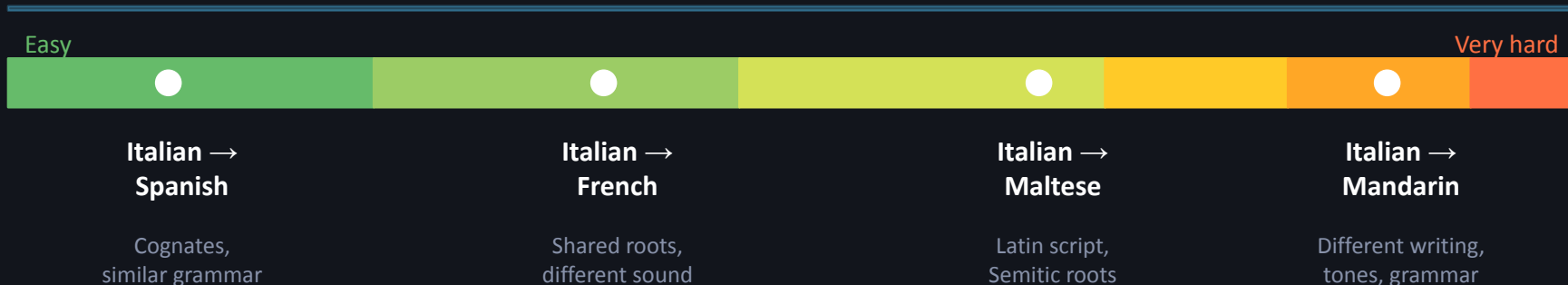
Train from scratch

- Needs ~1.28 M labelled images
- Three days on 8 GPUs
- Will not work with 2,000 examples

Transfer learning

- Start from a pretrained network
- Adapt to your task
- Minutes on a laptop CPU

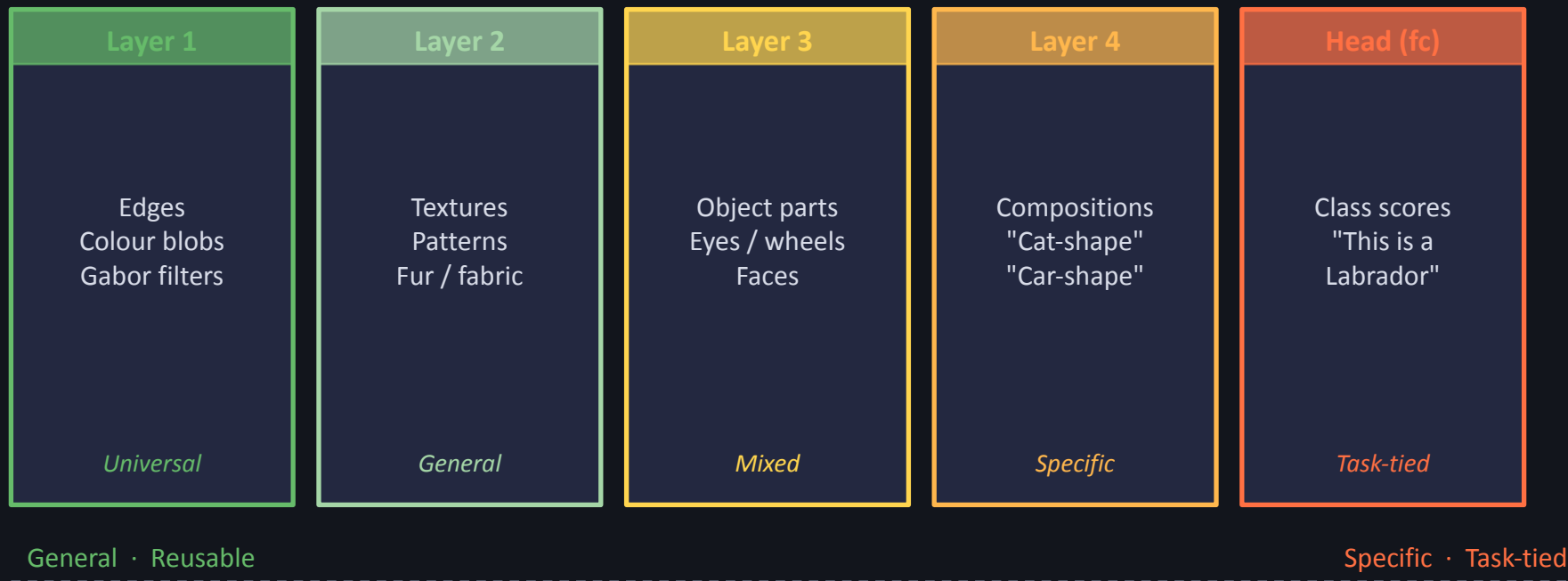
Analogy: Learning a New Language



The same gradient governs transfer learning

- Closer source ↔ target = more reusable knowledge = less fine-tuning needed
- Further apart = more layers need retraining = more data required
- Domain similarity is one of the two key considerations in the strategy decision
- Amount of Data is the second

What Each Layer Learns(Computer Vision Example)

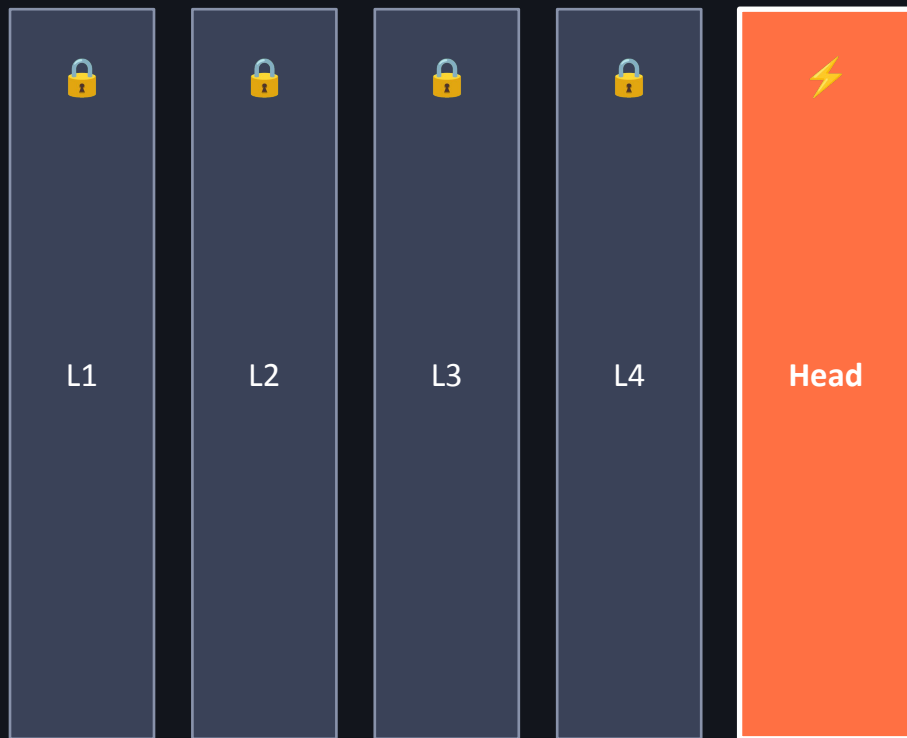


This hierarchy is emergent - not designed. Caused by receptive field growth + backprop gradient distribution.

Technique 1: Frozen Feature Extraction

- Backbone weights are fixed.
- Only the new head trains.

A linear classifier on top of pretrained features.



Backbone — frozen

Trains

Technique 2: End-to-End Fine-Tuning

- Unfreeze every layer.
- Train everything with a learning rate 10–100× lower than from-scratch training.
- Pretrained weights are nudged, not overwritten (hopefully).

Catastrophic forgetting

A high LR overwrites pretrained weights in the first few batches before the head has stabilised. The pretraining advantage is lost before it can be used.



← gradient flows backward through all layers

The Subtle Failure: LP-FT

Kumar et al. (2022): full fine-tuning improves in-distribution accuracy but often hurts out-of-distribution accuracy.

Naive fine-tuning

Random head + backbone train together from step 1

Random head sends chaotic gradients →
backbone features get distorted

-7% OOD accuracy on average

fix →

LP-FT

Stage 1: head only · Stage 2: everything

Stable head produces coherent gradients →
backbone adapts gracefully

Recovers OOD without losing ID accuracy

Kumar et al. (2022) — Fine-tuning can distort pretrained features and underperform from scratch

Technique 3: Gradual Unfreezing

- Start with only the head trainable.
- Unfreeze one more layer each stage, working backward from output to input.
- Each layer enters a stable environment.



→ Direction of unfreezing



Frozen



Trainable



Head (always on)

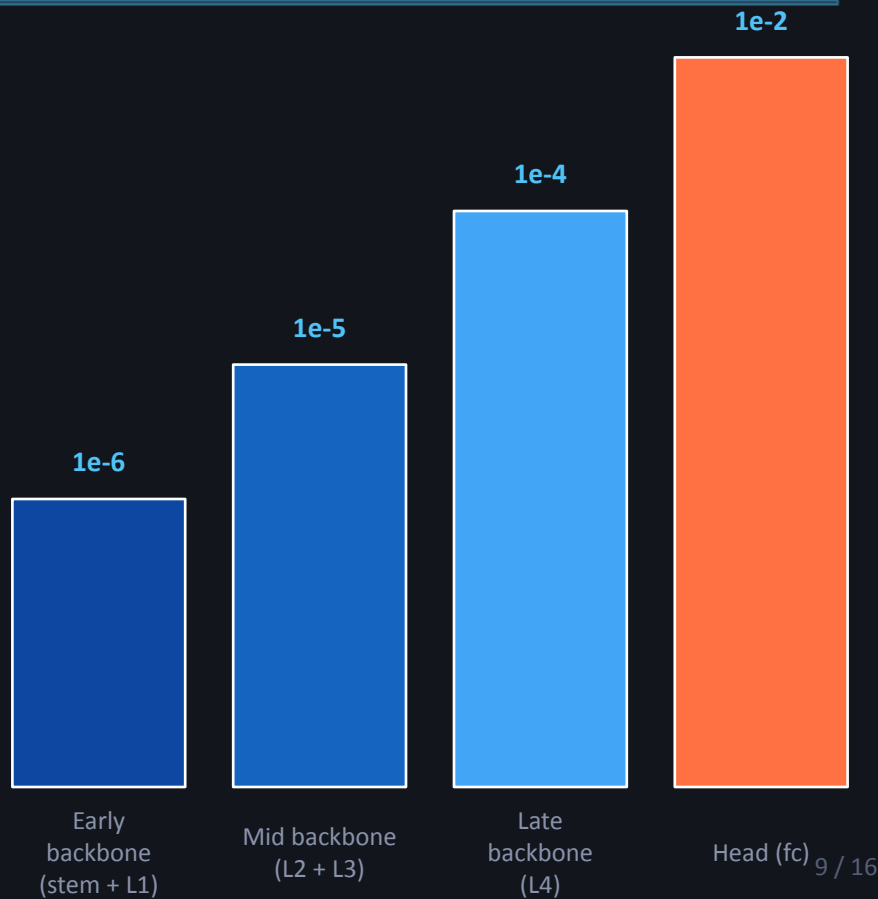
Technique 4: Discriminative Learning Rates

- Every layer trains, but each has its own learning rate.
- General layers change least.
- Specific layers change most.

$$\eta_l = \eta_L / \delta^{L-l}$$

$\delta \approx 2.6$ (NLP) or 10 (CV)

Complementary to gradual unfreezing - not an alternative.



Domain Shift

*The data your model meets at deployment differs from the data it was trained on.
This is why every technique on the previous slides exists.*

Covariate shift

Daylight photos → nighttime photos

Transfer learning/Fine-tuning fixes it

Label shift

50/50 spam dataset → 5% real spam

Reweighting / threshold adjustment/ transfer learning or fine-tuning might work

Concept shift

2010 spam patterns → 2024 phishing

*Hardest - usually needs new labels.
May require training from scratch*

When Each Technique Wins

Industrial defect detection

Few hundred labelled examples per class

Defects are visually distinct. Domain similar to ImageNet.

Frozen feature extraction

Linear probe on cached features

CheXNet — chest X-rays

100,000+ labelled X-rays · 14 disease classes

Domain different (grayscale medical), but large data.

Full end-to-end fine-tuning

DenseNet121 at low learning rate

Satellite imagery

Land-use segmentation · huge datasets

Top-down aerial view, often non-RGB. Very large target set.

Train from scratch

Or aggressive full fine-tune

Failure Cases & Limitations

Catastrophic forgetting

High LR overwrites pretrained weights before the head stabilises. All pretraining value is lost.

Fix: LR at $1e-4$ to $1e-5$ - 10–100× lower than from-scratch.

ID/OOD accuracy tradeoff

Aggressive uniform fine-tuning improves in-distribution accuracy but degrades OOD by ~7% on average.

Fix: LP-FT - stabilise head first, then unfreeze.

Pretraining doesn't always help

With a very large target dataset and drastically different domain, pre-training only speeds early convergence.

Fix: Consider training from scratch

When Not to Fine-Tune · Alternatives

Skip fine-tuning when...

- Frozen probe already meets your accuracy target
- Target dataset is huge and almost nothing like source
- Robustness to OOD matters more than peak ID accuracy

Alternatives

- Parameter-efficient methods (LoRA, adapters - ~1% of params)
- Few-shot / zero-shot prompting for tiny datasets
- Train from scratch - large data, very different domain

Quick Decision Recap

New task?

small + similar

**Frozen
extraction**

medium + any

**Gradual Unfreezing +
discriminative LRs**

large + different

**End-to-End Fine Tuning
OR
Train from scratch**

Universal practices

- Fine-tune at 10 to 100× lower LR than from-scratch
- Cache features when backbone is frozen (frozen feature extraction use mostly)
- Discriminative LRs: head largest, early backbone lowest
- Validate on OOD-like data before deployment
- When in doubt: start with the frozen probe baseline

Key References & Takeaways

Core papers

- Yosinski et al. (2014) - How transferable are features
- Howard & Ruder (2018) - ULMFiT
- Kornblith et al. (2019) - Do better ImageNet models transfer
- He et al. (2019) - Rethinking ImageNet pre-training
- Kumar et al. (2022) - Fine-tuning can distort features

Three things to remember

1. Layers learn general $\rightarrow$ specific features. Early layers transfer broadly whilst late layers don't.
2. Match technique to dataset size $\times$ domain similarity.
3. Fine-tune gently. Catastrophic forgetting is real.